

Technology Stack

Date	29 June 2026
Team ID	
Project Name	Rising Waters
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Technical Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Used to build responsive and user-friendly interface for entering data and displaying flood prediction results

2	Backend / Server-Side	Python, Flask	Flask is lightweight and integrates easily with machine learning models for handling prediction requests
3	Database / Data Storage	CSV dataset	Stores the flood dataset used for training and testing the prediction model. SQLite can store user inputs and prediction history
4	Cloud / Hosting / Deployment	Render	Provides free and easy deployment for flask applications with minimal configuration

5	Version Control & CI/CD	GitHub, Git	Tracks code changes, enables team collaboration, and securely stores the project repository
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Joblib	Pandas and NumPy process data, Scikit-learn trains the ML model, Matplotlib visualizes data, and joblib saves/loads the trained model